

Week 4 Evolutionary algorithm 2

1. What is selection pressure? How can we cause low pressure and how can we cause high pressure?
2. Why do we not always want high selection pressure?
3. Parent selection: what is fitness-proportionate selection, what is the problem with this type of selection? What is rank-based selection and how does this effect selection pressure? What is tournament selection?
4. Name some ways to choose the survivor selection? (3 ways)
5. Diversity preservation: What is the difference between implicit and explicit approaches for preserving diversity?
6. What is crowding and fitness sharing in explicit approach?
7. What is speciation and geographical separation in implicit approach?
8. What is hybridization and why do we do it?
9. What is memetic algorithm? How can we apply it to EA?
10. What is Lamarckian and Baldwinian in memetic algorithm?
11. What is multi-objective optimisation problems?
12. What two approaches do we have in MOP?
13. What is pareto front?